

Some Useful Commands

- Connect/Disconnect to/from system:

CONN[ECT] [[*logon*] [AS [SYSOPER|SYSDBA]]]

where *logon* requires the following syntax:

username[/password] [@net_service_name] | /

Connects a given username to Oracle. If you omit *net_service_name*, connects you to the default database. If you omit *username* and/or *password*, SQL*Plus prompts you for them. CONNECT followed by a slash (/) connects you using a default (OPSS) logon.

DISC[ONNECT]

Commits pending changes to the database and logs the current username off Oracle, but does not exit SQL*Plus.

- Connect/Disconnect to/from system:
- **sqlplus connect as sysdba**

```
PS C:\Users\DellUser> sqlplus connect as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Tue Feb 27 20:30:04 2024
Version 19.3.0.0.0

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Enter password:

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL>
```

Showing connection name

❖ `show con_name;`

```
SQL> show con_name;
```

```
CON_NAME
```

```
-----
```

```
CDB$ROOT
```

```
SQL>
```

Starting and Shutting Down a Database with SQL*Plus

startup command for starting up and *shutdown* for shutting it down

```
Enter password:
```

```
Connected to:
```

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> shutdown
```

```
Database closed.
```

```
Database dismounted.
```

```
ORACLE instance shut down.
```

```
SQL> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 2533356760 bytes
```

```
Fixed Size 9270488 bytes
```

```
Variable Size 587202560 bytes
```

```
Database Buffers 1929379840 bytes
```

```
Redo Buffers 7503872 bytes
```

```
Database mounted.
```

```
Database opened.
```

```
SQL>
```

STARTUP options

Some common startup options used with the STARTUP command in Oracle 19c,

1. PFILE=<path/filename>:

- Purpose: Specifies the location and filename of the initialization parameter file (PFILE) used to configure the database instance. This file contains various settings like memory allocation, network configuration, and storage parameters.
- Example: `STARTUP PFILE=/u01/app/oracle/product/19c/dbhome_1/dbs/initSID.ora`

2. SPFILE=<path/filename>:

- Purpose: Specifies the location and filename of the server parameter file (SPFILE) used to configure the database instance. This is a more secure way to store configuration parameters compared to the PFILE, as its contents are encrypted and not directly accessible.
- Example: `STARTUP SPFILE=/u01/app/oracle/product/19c/dbhome_1/dbs/spfileSID.ora` (assuming the SPFILE is named "spfileSID.ora")

STARTUP options

3. NOMOUNT:

- Purpose: Starts the instance without mounting any database files. This allows you to perform maintenance tasks on the instance itself, such as managing background processes or configuring network settings, without accessing the database data.
- Example: `STARTUP NOMOUNT`

4. CONTROL=<value>:

- Purpose: Specifies the level of control the instance exerts over automatic startup and shutdown. Common values are:
 - AUTO: The instance automatically attempts to start or stop based on defined parameters in the configuration files. This is often the default behavior.
 - MANUAL: The instance requires manual intervention (e.g., using the `STARTUP` or `SHUTDOWN` commands) to start or stop it.
- Example: `STARTUP CONTROL=MANUAL`

STARTUP options

5. RESTRICT:

- Purpose: Starts the instance in restricted mode, limiting access to only specific users or roles configured in the initialization parameter file. This is primarily used for maintenance purposes when access to the database should be restricted.

6. OPEN:

- Purpose: While not technically a startup option, it can be combined with other options like NOMOUNT to perform specific actions. For example, `STARTUP NOMOUNT OPEN` starts the instance without mounting the database but then immediately opens it for access.

SHUTDOWN options

- Some common shutdown options used with the SHUTDOWN command in Oracle 19c,

1. [NORMAL]:

- Purpose: This is the default behavior if no option is specified. It gracefully shuts down the database by waiting for all active connections to terminate, allowing ongoing transactions to complete, and then stopping database processes.
- Example: SHUTDOWN

2. IMMEDIATE:

- Purpose: This option terminates active connections and database processes immediately, potentially leading to data inconsistency if any transactions are ongoing. Use this option cautiously, as it can leave data in an incomplete state and require additional recovery steps during the next startup.
- Example: SHUTDOWN IMMEDIATE

SHUTDOWN options

3. ABORT:

- Purpose: This option abruptly terminates database processes immediately, leaving any transactions incomplete and potentially causing significant data inconsistencies. This option should be used as a last resort in critical situations and requires complete database recovery steps during the next startup.
- Example: SHUTDOWN ABORT

4. FINAL [=n]:

- Purpose: This option performs a final shutdown, waiting for ongoing recovery processes to complete before stopping the instance. The optional n parameter specifies the maximum number of seconds to wait for recovery to finish before forcing the shutdown.
- Example: SHUTDOWN FINAL or SHUTDOWN FINAL=300 (wait up to 300 seconds for recovery)

SHUTDOWN options

5. TRANSACTION WAIT:

- Purpose: This option waits for all ongoing transactions to complete before shutting down the database. It behaves similarly to the normal shutdown but might take longer depending on the number of active transactions.
- Example: SHUTDOWN TRANSACTION WAIT

Listing CDB and PDB informations

- **V\$CONTAINERS:** This view displays details about all containers (CDB and PDBs) in the instance.

```
SELECT NAME, CON_ID, OPEN_MODE  
FROM V$CONTAINERS;
```

```
SQL> SELECT NAME, CON_ID, OPEN_MODE  
2 FROM V$CONTAINERS;
```

```
NAME
```

```
-----  
CON_ID OPEN_MODE  
-----
```

```
CDB$ROOT
```

```
1 READ WRITE
```

```
PDB$SEED
```

```
2 READ ONLY
```

```
ORCLPDB
```

```
3 MOUNTED
```

```
SQL>
```

Listing CDB and PDB informations

- **V\$PDBS:** This view provides information about PDBs, including their name, status (open_mode), and CDB association:

```
SELECT NAME, OPEN_MODE, CON_ID  
FROM V$PDBS;
```

```
SQL> SELECT NAME, OPEN_MODE, CON_ID  
2 FROM V$PDBS;
```

```
NAME
```

```
-----  
OPEN_MODE      CON_ID  
-----
```

```
PDB$SEED
```

```
READ ONLY      2
```

```
ORCLPDB
```

```
MOUNTED        3
```

```
SQL>
```

Connecting to PDB

- ❑ Connect to CDB
- ❑ Use the ALTER SESSION command with the CONTAINER clause to switch to a specific PDB:

ALTER SESSION SET CONTAINER = orclpdb;

NOTE: switching between PDBs within the CDB requires at least SYSDBA or SYSOPER privileges.

Connecting to PDB

```
Enter user-name: sys as sysdba
Enter password:

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> ALTER SESSION SET CONTAINER=orclpdb;

Session altered.

SQL> show con_name;

CON_NAME
-----
ORCLPDB
SQL>
```

Creating a stored procedure to display message and executing

Here is an example of a simple PL/SQL stored procedure that displays a "Hello, World!" message:

```
CREATE OR REPLACE PROCEDURE display_hello_world IS
BEGIN
    DBMS_OUTPUT.PUT_LINE('Hello, World!');
END display_hello_world;
/
```

```
SET SERVEROUTPUT ON
EXEC display_hello_world;
```

Oracle Listener control commands

- A listener is a process that listens for incoming client connection requests and manages the traffic to the database.
- It plays a crucial role in establishing and maintaining communication between database clients and the Oracle database server.
- The listener is configured through a file called listener.ora.
- The **lsnrctl** command-line utility is used to control and manage the Oracle Listener. You can use it to start, stop, and configure the listener.

Oracle Listener control commands

- . Some commonly used commands include:
 - **lsnrctl start:** Starts the listener.
 - **lsnrctl stop:** Stops the listener.
 - **lsnrctl status:** Displays the current status and configuration of the listener.
 - **lsnrctl reload:** Reloads the listener configuration without stopping it.
- The listener is responsible for knowing which services (databases or instances) it should be listening for. Database services are typically registered with the listener in the database configuration.

Unlocking HR-Schema in Oracle 19 C

A **schema** is a collection of logical structures of data, or schema objects. Furthermore a schema is owned by a database user and has the same name as that of the user. Also, each user owns a single schema.

Unlocking sample accounts like HR in Oracle 19c is a three-step process. These are –

1. Create A TNS service

[use NetCA or manually update tnsnames.ora then reload listener by using 'lsnrctl reload' command from OS administrative level]

2. Switch the Container [to ORCLPDB]
3. Unlock the Account.

```
ALTER USER hr IDENTIFIED BY <your_pass> ACCOUNT UNLOCK;
```

Viewing user, account_status and default Tablespaces

```
select username, account_status,default_tablespace from dba_users;
```

```
SQL> select username, account_status,default_tablespace from dba_users;
```

USERNAME		
-----	-----	-----
ACCOUNT_STATUS		DEFAULT_TABLESPACE
-----	-----	-----
SYS		
OPEN		SYSTEM
SYSTEM		
OPEN		SYSTEM
XS\$NULL		
EXPIRED & LOCKED		SYSTEM

Displaying Current User in the system

```
SQL> show user;  
USER is "SYS"  
SQL>
```

Displaying Privilege granted to a user

Command :

```
select privilege from dba_sys_privs where grantee='HR';
```

```
SQL> select privilege from dba_sys_privs where grantee='HR';
```

```
PRIVILEGE
```

```
-----
```

```
CREATE DATABASE LINK
```

```
ALTER SESSION
```

```
CREATE VIEW
```

```
UNLIMITED TABLESPACE
```

```
CREATE SESSION
```

```
CREATE SEQUENCE
```

```
CREATE SYNONYM
```

```
7 rows selected.
```

```
SQL>
```

Displaying all the users in the database

```
select username from dba_users;
```

```
USERNAME
```

```
-----
```

```
SYS
```

```
SYSTEM
```

```
XS$NULL
```

```
LBACSYS
```

```
OUTLN
```

```
DBSNMP
```

```
APPQOSSYS
```

```
DBSFUSER
```

```
GGSYS
```

```
ANONYMOUS
```

```
UP
```

Formatting output in the SQL Plus: Setting linesize, formatting column and limiting number of results

```
SQL> set linesize 900;
```

```
SQL> column username format a30;
```

```
SQL> column default_tablespace format a30;
```

```
SQL> column account_status format a30;
```

```
SQL> select username, account_status, default_tablespace from dba_users where rownum<12;
```

Formatting output in the SQL Plus

```
SQL> select username, account_status,default_tablespace from dba_users where rownum<12;
```

USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
SYS	OPEN	SYSTEM
SYSTEM	OPEN	SYSTEM
XS\$NULL	EXPIRED & LOCKED	SYSTEM
LBACSYS	LOCKED	SYSTEM
OUTLN	LOCKED	SYSTEM
DBSNMP	LOCKED	SYSAUX
APPQOSSYS	LOCKED	SYSAUX
DBSFUSER	LOCKED	SYSAUX
GGSYS	LOCKED	SYSAUX
ANONYMOUS	EXPIRED & LOCKED	SYSAUX
HR	LOCKED	SYSAUX

```
11 rows selected.
```

Displaying Tables in a Schema:

```
SELECT table_name FROM dba_tables WHERE owner = 'your_schema_name';
```

```
SQL> SELECT table_name  
2 FROM dba_tables  
3 WHERE owner = 'HR';
```

```
TABLE_NAME  
-----
```

```
REGIONS  
LOCATIONS  
DEPARTMENTS  
JOBS  
EMPLOYEES  
JOB_HISTORY  
COUNTRIES
```

Viewing Tablespace Information:

DBA_TABLESPACES view:

This view offers comprehensive information about all tablespaces in the database.

```
SELECT tablespace_name, status, block_size/1024 as  
block_size_kb, max_size/1024/1024/1024 as max_size_gb FROM  
dba_tablespaces;
```

TABLESPACE_NAME	STATUS	BLOCK_SIZE_KB	MAX_SIZE_GB
SYSTEM	ONLINE	8	2
SYSAUX	ONLINE	8	2
UNDOTBS1	ONLINE	8	2
TEMP	ONLINE	8	2
USERS	ONLINE	8	2
SQL>			

Viewing Tablespace Information

USER_TABLESPACES view:

This view shows tablespaces accessible to the current user

```
SELECT tablespace_name, status FROM  
user_tablespaces;
```

```
SQL> connect hr@orclpdb;  
Enter password:  
Connected.  
SQL> SELECT tablespace_name, status  
2 FROM user_tablespaces;
```

TABLESPACE_NAME	STATUS
SYSTEM	ONLINE
SYSAUX	ONLINE
UNDOTBS1	ONLINE
TEMP	ONLINE
USERS	ONLINE

```
SQL>
```

Viewing Tablespace Information

V\$TABLESPACE View:

This view belongs to the category of dynamic performance views (V\$ views). It provides real-time information about the current state of tablespaces in the database. However, it offers a more concise set of details compared to DBA_TABLESPACES.

```
SQL> SELECT * from v$tablespace;
```

TS#	NAME	INC	BIG	FLA	ENC	CON_ID
0	SYSTEM	YES	NO	YES		3
1	SYSAUX	YES	NO	YES		3
2	UNDOTBS1	YES	NO	YES		3
3	TEMP	NO	NO	YES		3
5	USERS	YES	NO	YES		3

```
SQL>
```

Knowing the type of tablespace

- To determine the type of a tablespace in Oracle, we can use the V\$TABLESPACE or DBA_TABLESPACES view.
- Both views contain a column named CONTENTS that indicates the type. Here's how you can use this information:

```
SELECT tablespace_name, contents FROM dba_tablespaces;
```

```
SQL> SELECT tablespace_name, contents FROM dba_tablespaces;
```

TABLESPACE_NAME	CONTENTS
-----	-----
SYSTEM	PERMANENT
SYSAUX	PERMANENT
UNDOTBS1	UNDO
TEMP	TEMPORARY
USERS	PERMANENT

```
SQL>
```

Creating tablespace

Basic syntax of CREATE TABLESPACE to create a tablespace

CREATE TABLESPACE tablespace_name

DATAFILE 'datafile_path'

SIZE [size_tablespace];

```
CREATE TABLESPACE bipin_tbs1 DATAFILE 'bipin_df1.dat' SIZE 2M;
```

★ This creates permanent tablespace.

Creating tablespace

```
CON_NAME
-----
ORCLPDB
SQL> CREATE TABLESPACE bipin_tbs1 DATAFILE 'bipin_df1.dat' SIZE 2M;

Tablespace created.

SQL> SELECT tablespace_name, contents, max_size/1024/1024 as max_size_mb FROM dba_tablespaces;
```

TABLESPACE_NAME	CONTENTS	MAX_SIZE_MB
-----	-----	-----
SYSTEM	PERMANENT	2048
SYSAUX	PERMANENT	2048
UNDOTBS1	UNDO	2048
TEMP	TEMPORARY	2048
USERS	PERMANENT	2048
BIPIN_TBS1	PERMANENT	2048

```
6 rows selected.
```

Alter Tablespace

- Use the ALTER TABLESPACE statement to alter an existing tablespace or one or more of its data files or temp files.
- To change the status to offline

```
ALTER tablespace bipin_tbs1 OFFLINE;
```

- To change the status to online

```
ALTER tablespace bipin_tbs1 ONLINE;
```

- To add a data file

```
ALTER TABLESPACE bipin_tbs1 ADD DATAFILE 'bipin_df.dat' SIZE 3M;
```

- To rename

```
ALTER TABLESPACE old_tablespace_name RENAME TO new_tablespace_name;
```

```
SQL> ALTER tablespace bipin_tbs1 OFFLINE;
```

```
Tablespace altered.
```

```
SQL> SELECT tablespace_name, contents,status FROM dba_tablespaces;
```

TABSPACE_NAME	CONTENTS	STATUS
SYSTEM	PERMANENT	ONLINE
SYSAUX	PERMANENT	ONLINE
UNDOTBS1	UNDO	ONLINE
TEMP	TEMPORARY	ONLINE
USERS	PERMANENT	ONLINE
BIPIN_TBS1	PERMANENT	OFFLINE

```
6 rows selected.
```

```
SQL>
```

```
SQL> ALTER tablespace bipin_tbs1 ONLINE;
```

```
Tablespace altered.
```

```
SQL> ALTER TABLESPACE bipin_tbs1 ADD DATAFILE 'bipin_df.dat' SIZE 3M;
```

```
Tablespace altered.
```

```
SQL> ALTER TABLESPACE bipin_tbs2 RENAME TO renamed_bipin_tbs2;
```

```
Tablespace altered.
```

```
SQL> ALTER TABLESPACE renamed_bipin_tbs2 RENAME TO bipin_tbs2;
```

```
Tablespace altered.
```

Deleting Tablespace

- Use the **DROP TABLESPACE** statement to remove a tablespace from the database.
- When you drop a tablespace, Oracle Database does not place it in the recycle bin. Therefore, you cannot subsequently either purge or undrop the tablespace.

We can use the DROP TABLESPACE statement to permanently remove it from the database:

```
DROP TABLESPACE bipin_tbs1;
```

To drop the tbs_02 tablespace and deletes all associated operating system data files:

```
DROP TABLESPACE tbs_02 INCLUDING CONTENTS AND DATAFILES;
```

Deleting Tablespace

```
SQL> DROP TABLESPACE bipin_tbs1;
```

```
Tablespace dropped.
```

```
SQL> SELECT tablespace_name, contents FROM dba_tablespaces;
```

TABLESPACE_NAME	CONTENTS
SYSTEM	PERMANENT
SYSAUX	PERMANENT
UNDOTBS1	UNDO
TEMP	TEMPORARY
USERS	PERMANENT

```
SQL>
```

Getting information about data files

- We can obtain information about datafiles in Oracle Database 19c using the **DBA_DATA_FILES** or **V\$DATAFILE** views.
- The DBA_DATA_FILES view provides details about datafiles and is typically accessible to users with DBA privileges.
- The V\$DATAFILE dynamic performance view offers similar information and is available to users with the necessary privileges.

```
SELECT tablespace_name, file_name, bytes/(1024*1024) AS  
MB_SIZE, online_status FROM DBA_DATA_FILES;
```

```
SELECT file#, name, bytes/1024/1024 as size_in_mb,  
status, con_id FROM V$DATAFILE;
```

Getting information about data files

TABLESPACE_NAME	FILE_NAME	MB_SIZE	ONLINE_STATUS
SYSTEM	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF	270	SYSTEM
SYSAUX	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\SYSAUX01.DBF	310	ONLINE
UNDOTBS1	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF	100	ONLINE
USERS	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\USERS01.DBF	5	ONLINE
BIPIN_TBS2	C:\ORACLE19C_HOME\DATABASE\BIPIN_DF3.DAT	4	ONLINE

SQL>

Getting information about datafiles

```
SQL> SELECT file#, name, bytes/1024/1024 as size_in_mb, status, con_id FROM V$DATAFILE;
```

FILE#	NAME	SIZE_IN_MB	STATUS	CON_ID
1	D:\APP\DELLUSER\ORADATA\ORCL\SYSTEM01.DBF	900	SYSTEM	1
3	D:\APP\DELLUSER\ORADATA\ORCL\SYSAUX01.DBF	630	ONLINE	1
4	D:\APP\DELLUSER\ORADATA\ORCL\UNDOTBS01.DBF	65	ONLINE	1
5	D:\APP\DELLUSER\ORADATA\ORCL\PDBSEED\SYSTEM01.DBF	260	SYSTEM	2
6	D:\APP\DELLUSER\ORADATA\ORCL\PDBSEED\SYSAUX01.DBF	280	ONLINE	2
7	D:\APP\DELLUSER\ORADATA\ORCL\USERS01.DBF	5	ONLINE	1
8	D:\APP\DELLUSER\ORADATA\ORCL\PDBSEED\UNDOTBS01.DBF	100	ONLINE	2
9	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF	270	SYSTEM	3
10	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\SYSAUX01.DBF	310	ONLINE	3
11	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF	100	ONLINE	3
12	D:\APP\DELLUSER\ORADATA\ORCL\ORCLPDB\USERS01.DBF	5	ONLINE	3
FILE#	NAME	SIZE_IN_MB	STATUS	CON_ID
15	C:\ORACLE19C_HOME\DATABASE\BIPIN_DF3.DAT	4	ONLINE	3

12 rows selected.

Renaming, Resizing and dropping datafiles

- CREATE TABLESPACE statement **creates datafiles** specified.
- ALTER TABLESPACE statement can be used to modify attributes of an existing tablespace, such as **adding datafiles, resizing datafiles**, or changing storage parameters.
 - You can add the AUTOEXTEND ON MAXSIZE size clause to enable autoextension for the datafile. This allows the datafile to grow automatically when it reaches its initial size limit.
- The DROP TABLESPACE statement **removes** an entire tablespace, including its **datafiles** and all contained objects.
- The **ALTER DATABASE DATAFILE** statement is used for specific operations on individual datafiles, such as renaming or relocating a datafile.

Renaming, Resizing and dropping datafiles

Resizing Using ALTER DATABASE:

`ALTER DATABASE DATAFILE 'file_name' RESIZE new_size;`

Example: `ALTER DATABASE DATAFILE 'bipin_df3.dat' RESIZE 200M`

```
; SQL> ALTER DATABASE DATAFILE 'bipin_df3.dat' RESIZE 200M;  
Database altered.
```

Renaming, Resizing and dropping datafiles

To rename the datafile use following statement

```
ALTER DATABASE RENAME FILE 'old_file_name' TO 'new_file_name';
```

Use the **ALTER TABLESPACE DROP DATAFILE '<datafile_path>'** statement to drop the datafile from the tablespace.

→ It involves dropping the datafile itself while keeping the tablespace.

These statements needs additional settings !!

Creating user in pdb

- First open database
 - ALTER DATABASE OPEN
- Create user
 - CREATE USER username IDENTIFIED BY password

```
SQL> ALTER SESSION SET CONTAINER = ORCLPDB;  
Session altered.  
SQL> ALTER DATABASE OPEN;  
Database altered.  
SQL> CREATE USER MEBPIN IDENTIFIED BY MYPASSWORD123;  
User created.  
SQL>
```

Granting connect privilege to user

- GRANT CONNECT TO username;

```
SQL> GRANT CONNECT TO MEBPIN;  
  
Grant succeeded.  
  
SQL>
```

Connecting with new user

Error!

```
Enter user-name: mebpin@orclpdb;  
Enter password:  
ERROR:  
ORA-12154: TNS:could not resolve the connect identifier specified
```